# A Trusted Generative-Discriminative Joint Feature Learning Framework for Remote Sensing Image Classification

Lingyu Si, Hongwei Dong, Wenwen Qiang, Zeen Song, Bo Du, *Senior Member, IEEE*, Junzhi Yu, *Fellow, IEEE*, and Fuchun Sun, *Fellow, IEEE*

***Abstract*— Remote sensing image (RSI) classification is a popular research topic that aims to assign semantic labels to images acquired from aerial or maritime platforms. Existing deep feature learning methods for this task can be divided into two paradigms: generative and discriminative. The former methods are good at capturing every local detail of images, while the later approaches focus on the most salient area. The significant differences between the two types of methods, both in terms of their underlying mechanisms and practical implementation, motivate us to integrate information acquired by both paradigms by exploiting their complementary strengths. However, this idea faces a challenge that local information in the extracted features, especially those from generative methods, may not be reliable for RSI classification. The reason for this challenge is that, due to the characteristics of the ground observation perspective, some RSIs, while semantically different, exhibit a significant degree of similarity in local details. This phenomenon leads to insufficient discriminability of local features to separate multiple RSI categories, which implies that the classification results overly focused on local information may be unreliable. To address this issue, in this article, we propose a novel framework that integrates generative and discriminative feature learning methods with evidential learning for RSI classification. Our framework uses the Dirichlet distribution to model the predicted probabilities to be integrated, thereby collecting evidence about their reliability. This enables us to integrate multiple features at an evidence level and make reliable decisions, overcoming the unreliabilities of generative-discriminative joint feature learning induced by RSI characteristics. We evaluate the proposed framework on several satellite and shipborne RSI classification datasets. The experimental results show that our method outperforms the state-of-the-art baselines in terms of accuracy and robustness.**

***Index Terms*— Deep learning, evidential learning, generative feature learning, image classification, remote sensing.**

Manuscript received 7 December 2023; accepted 11 December 2023. Date of publication 13 December 2023; date of current version 21 December 2023. This work was supported in part by the National Natural Science Foundation of China under 62301539; and in part by the China Postdoctoral Science Foundation under Grant 2023M733615, Grant 2023M743639, and Grant E3YD5901. *(Lingyu Si and Hongwei Dong contributed equally to this work.) (Corresponding authors: Wenwen Qiang; Hongwei Dong.)*

Lingyu Si, Wenwen Qiang, and Zeen Song are with the Science and Technology on Integrated Information System Laboratory, Institute of Software, Chinese Academy of Sciences, Beijing 100191, China (e-mail: lingyu@iscas.ac.cn; qiangwenwen@iscas.ac.cn; songzeen@iscas.ac.cn).

Hongwei Dong is with the Science and Technology on Integrated Information System Laboratory, Institute of Software, Chinese Academy of Sciences, Beijing 100191, China, and also with the Department of Information Engineering, Harbin Institute of Technology, Harbin 150001, China (e-mail: donghongwei1994@163.com).

Bo Du is with the Hubei Luojia Laboratory, National Engineering Research Center for Multimedia Software, School of Computer Science, Wuhan University, Wuhan 430079, China (e-mail: dubo@whu.edu.cn).

Junzhi Yu is with the College of Engineering, Peking University, Beijing 100871, China (e-mail: yujunzhi@pku.edu.cn).

Fuchun Sun is with the Department of Computer Science and Technology, Tsinghua University, Beijing 100190, China (e-mail: fcsun@mail.tsinghua.edu.cn).

Digital Object Identifier 10.1109/TGRS.2023.3342740

## I. Introduction

THE rapid development of remote sensing technology has led to the availability of large-scale and high-resolution remote sensing images (RSIs). These images can be used for various applications, such as information mining [1], retrieving similar images [2], and generating natural language descriptions [3]. A key prerequisite for these applications is RSI classification [4], which aims to assign semantic labels to each patch or pixel in an RSI based on its target type.

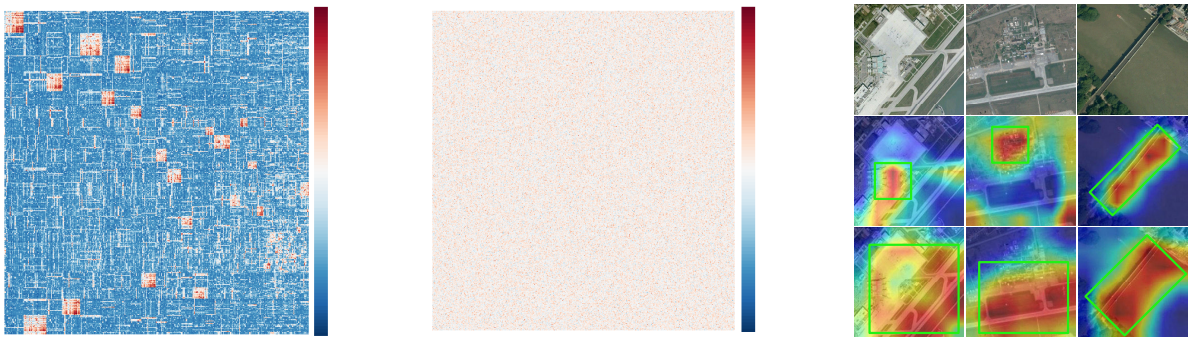
RSI classification is challenging due to the presence of many ambiguous targets that share similar visual characteristics. To achieve accurate classification, it is essential to extract the features that can capture the variations between different categories and the similarities within the same category. In the literature, two kinds of features have been proposed for RSI classification: hand-crafted features and data-driven features. Hand-crafted features are designed based on prior knowledge and domain expertise, while data-driven features are learned automatically from the data by deep neural networks [5], [6]. Data-driven features are generally more nonlinear and abstract than hand-crafted features and have shown superior performance in various RSI classification tasks [7], [8].

Generative and discriminative methods are two types of data-driven feature learning methods that differ in their learning objectives [9]. Generative methods, such as deep autoencoder (DAE) [10], aim to reconstruct inputs with high fidelity by encoding and decoding them without supervision, which requires capturing both global semantics and local details from the images. Discriminative methods, such as convolutional neural network (CNN) [11], aim to make predictions with high accuracy by minimizing the empirical error between the predicted probabilities and the ground truth labels with supervision. This requires extracting the most salient feature semantics that match the true labels.

Generative and discriminative feature learning methods are both state-of-the-art approaches for RSI classification, but their potential synergy has not been investigated in the field. Our hypothesis is that *these methods have complementary strengths*

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:12:10 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c)

Fig. 1. Results of our motivating experiments. (a) Similarity between generative and discriminative features with hierarchical clustering. (b) Similarity between the eigenvectors of generative and discriminative features. (c) Source image (top row), attention distribution visualization of generative (bottom row) and discriminative (middle row) models. (a) and (b) Colors from red to blue represent values of similarity ranging from 1 to −1, meaning that the variables to be compared range from exactly the same to completely different. Therefore, a smaller red area indicates a lower similarity between the two sets of features. (c) Colors from red to blue in the heatmap indicate a decrease in attention from high to low. Therefore, a larger red area indicates that the model attends to a greater number of regions, thereby reducing the likelihood of overlooking local details.

*due to their different learning objectives and feature extraction mechanisms*. Certainly, significant differences among generative and discriminative features constitute a basic premise for the establishment of this hypothesis. To validate this assertion, we employ the same RSI dataset to train generative and discriminative models independently and use cosine similarity [$\cos(\boldsymbol{a}, \boldsymbol{b}) = \boldsymbol{ab}/|\boldsymbol{a}||\boldsymbol{b}|$] to quantitatively assess the degree of their feature similarity. Fig. 1(a) and (b) shows the experimental results, where the feature similarity is calculated twice to ensure a comprehensive comparison: First, by assessing the cosine similarity matrix among each dimension of the original features (with hierarchical clustering [12] for better visual effect). Second, by assessing the cosine similarity matrix between corresponding eigenvectors obtained by principal component analysis. Analyzing the results according to the explanation in the caption, it becomes apparent that there are significant differences in the numerical values of the extracted features between the two models, which is consistent with Xie et al. [13] and provides a basic premise for the integration of generative and discriminative feature learning methods. Building upon the established numerical differences between the two types of features, we further employ class activation mapping (CAM) [14] to measure feature attention patterns, aiming to demonstrate the existence of complementary strengths between generative and discriminative features. The results, presented in the form of heatmaps in Fig. 1 (c), visually illustrate the noticeable differences in attention distributions between generative and discriminative features. The former exhibits a larger hot zone, indicating a higher focus on local information in its features, while the characteristics manifested by the latter are precisely the opposite. Given the significance of local information for RSI image classification, this observation establishes a foundation for the complementary strengths between generative and discriminative learning methods.

Motivated by the above-mentioned findings, we aim to construct a novel framework that integrates generative and discriminative feature learning methods in a joint learning manner. The key idea is to leverage the information captured by the generative feature learning method as a complementary to the discriminative task, thus enhancing the RSI classification performance [15], [16]. However, this idea poses a significant challenge: *how to deal with the unreliability of local information in RSI classification?* Before presenting our solution, we first explain the source of this unreliability using Fig. 2.

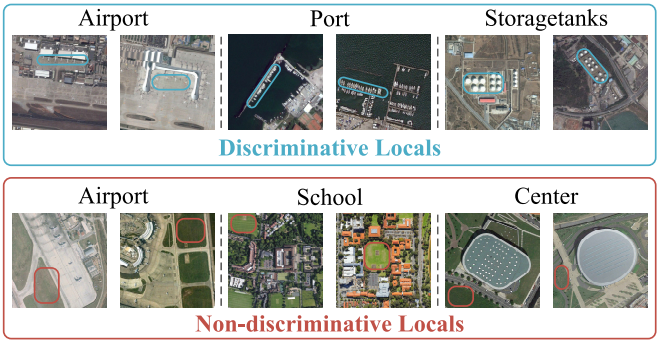



Fig. 2. Some examples to illustrate the unreliability of local information in RSI classification. In the upper part, the local details of aircraft, ships, and storage tanks (marked in blue) are the key information to distinguish the images belonging to the corresponding categories. However, in the lower part, the local detail of grass (marked in red) is shared by images from different categories, such as aircraft, schools, and centers. Therefore, it becomes nondiscriminative and may confuse the classifier.

In many cases, local features play a crucial role in semantic prediction, as illustrated in the upper part of Fig. 2. However, RSIs possess significantly larger perceptual distances and real scene sizes per pixel compared with natural images. This implies that multiple categories in RSIs can share similar local details, making them nondiscriminative for classification. For instance, as depicted in the lower part of Fig. 2, grass serves as a shared local detail among the categories of airport, school, and center. Relying on this local information can lead to model overfitting and unreliable predictions, as the features of grass are not discriminative enough to separate these categories. Therefore, it becomes essential to assess the reliability of information and ensure trusted integration during generative-discriminative joint learning, considering the undifferentiated attention given to every local detail in generative models.

To address this issue, in this article, we introduce evidential learning to achieve the trusted integration of generative and discriminative feature learning methods for RSI classification. In the proposed learning framework, we use an evidence-parameterized Dirichlet distribution to quantify the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:12:10 UTC from IEEE Xplore. Restrictions apply.

uncertainty of the predicted probabilities of generative and discriminative feature learning models, thus performing the integration of their features and decisions at an evidence level. Therefore, we can reduce the influence of nondiscriminative features on the classification results and overcome the unreliability problem in generative-discriminative joint feature learning caused by RSI characteristics. In summary, the main contributions of this article are threefold.

1) We highlight a key but neglected issue in RSI classification, i.e., the complementarity between generative and discriminative feature learning methods, and empirically demonstrate the feasibility of this issue.
2) We propose a trusted generative-discriminative joint feature learning framework that can integrate multiple types of features and decisions in a reliable way, providing an effective and model-agnostic learning paradigm for plug-and-play improvements of RSI classification performance.
3) We evaluate our proposed framework on several RSI classification datasets from different sources, including airborne and shipborne, and show that it outperforms existing algorithms in terms of accuracy and robustness.

## II. Related Works

This section provides a comprehensive review of the research literature related to this article, including data-driven features for RSI classification and evidential deep learning.

### A. Data-Driven Features for RSI Classification

Depending on the learning paradigm, data-driven feature learning methods for RSI classification can be categorized into the following two types.

*1) Generative Feature Learning Methods:* Generative feature learning methods, such as DAEs [17] and generative adversarial networks (GANs) [18], are trained to reconstruct input data or learn its underlying distribution. These methods can generate new data that is similar to the input, as well as extract meaningful and informative features from the data without any supervision. The features can be used for downstream tasks, such as classification, clustering, and so on.

As one of the most popular methods for generative feature learning, DAEs consist of two components: an encoder and a decoder. The encoder compresses the data from a higher dimensional space to a lower dimensional latent space, which can be seen as a feature representation of the data. The decoder reconstructs the original data from the latent space. The encoder and decoder are trained jointly to minimize the reconstruction error, which is usually measured by mean squared error or binary cross-entropy.

DAEs have been widely used in the RSI field, including but not limited to very-high-resolution (VHR) [19], multispectral [20], hyperspectral [21], and radar [22] images. Various types of DAEs have been proposed and applied to RSI classification. Zhang et al. [23] proposed a DAE-based unsupervised feature learning framework for VHR images. Chen et al. [24] proposed a data-driven feature extraction method for hyperspectral image land cover classification with DAEs. They proved that useful high-level features can be extracted by the DAE model. Geng et al. [25] improved the conventional DAEs with the convolutional operation to introduce more spatial information. They have shown that the extracted features are robust to noise and perform well for synthetic aperture radar image classification. Besides, sparse [26], denoising [27], contractive [28], and variational [29] DAEs have also been studied and have shown good performance in extracting robust and discriminative features for RSI classification.

*2) Discriminative Feature Learning Methods:* Discriminative feature learning methods are techniques that directly use deep neural networks to learn features that can distinguish between different data categories. They typically employ an end-to-end network that minimizes the loss function between the prediction and the ground truth label, achieving supervised training and mining discriminative features.

One of the most prominent discriminative feature learning methods is CNN [30], which has shown excellent performance in 2-D image data processing. A typical CNN consists of multiple layers, including convolution, pooling, and fully connected layers. The convolution layers apply a set of learnable filters to the input image, performing local feature extraction. The pooling layers downsample the feature maps generated by the convolution layers, reducing the spatial dimension while preserving the most salient features. The fully connected layers learn the class-specific representations and make preparations for the final classification decisions by combining the high-level features extracted from the previous layers. The stacking of multiple convolution and pooling layers implements the process of aggregation from local information to global information.

CNNs have been widely applied to the RSI field and have achieved remarkable results. Penatti et al. [31] and Marmanis et al. [32] introduced CNNs as fixed feature extractors for RSI scene classification and demonstrated their superior generalization ability over low-level hand-crafted features. However, subsequent studies have shown that fine-tuning the pretrained CNN models performs better than using the fixed feature extractors [33]. Cheng et al. [34] improved conventional CNNs by adding a metric learning-based regularization term for learning features with better discriminability from VHR RSIs. Wang et al. [35] proposed an end-to-end attention recurrent convolutional network for RSI scene classification. Moreover, CNNs also exhibited the state-of-the-art performance in the classification of hyperspectral [36] and radar images [37].

### B. Evidential Deep Learning

The softmax function is widely used in discriminative feature learning methods to convert the continuous activation of the output layer into predicted probabilities. However, the softmax function has a drawback: it tends to exaggerate the predicted probabilities by squashing the ratios of predicted probabilities into a simplex that fits the one-hot encoded ground truth labels. As a result, the classification probability calculated using the softmax function can be high even for misclassified samples.

To address this problem, uncertainty-based learning methods are proposed to learn and express uncertainty using neural networks. These methods can be divided into two categories:

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:12:10 UTC from IEEE Xplore. Restrictions apply.

Bayesian and non-Bayesian. Bayesian methods characterize uncertainty by learning distributions over model weights [38], but they incur high computational costs. Non-Bayesian methods avoid explicitly learning the distributions and use various techniques to model uncertainty [39], [40], [41].

Evidential deep learning is a non-Bayesian method that leverages subjective logic theory [42] to directly model uncertainty. Subjective logic is a generalization of probabilistic logic that handles both aleatoric and epistemic uncertainty using belief functions. Within this framework, one can quantify belief masses and estimate the uncertainty of a deep neural network through the Dirichlet distribution [43]. To measure uncertainty under a single model or view, conventional evidential learning methods are often used. However, these methods cannot handle the integration of multiple models or views at an evidence level. To address this challenge, some researchers have proposed methods based on the Dempster–Shafer theory [44], [45] and opinion aggregation [46].

## III. Methodology

This section presents a detailed introduction to our proposed trusted generative-discriminative joint feature learning framework. The paradigms of generative and discriminative feature learning methods are introduced first, followed by the implementation of the integration strategy, including the uncertainty quantification and evidence integration.

### A. Generative Feature Learning

We consider a supervised learning setting where the input dataset consists of $N$ pairs of training samples and their corresponding labels, denoted by $X = \{(x_i, \boldsymbol{y}_i)\}_{i=1}^{i=N}$, where $x_i$ is an image and $\boldsymbol{y}_i$ is the vector of one-hot encoded ground truth label. To learn generative features from the data, we employ a convolutional autoencoder (CAE) as our generative feature learning model. A CAE consists of two mapping functions: an encoder $f_g^{\text{en}}(\cdot)$ that transforms the input into a latent representation using convolution operations, and a decoder $f_g^{\text{de}}(\cdot)$ that reconstructs the input from the latent representation using transposed convolution operations. Although we focus on CAEs in this work, our approach can be easily generalized to other types of generative models, such as variational autoencoders or GANs.

Given an input RSI image of size $w \times h \times c_{\text{in}}$, an encoder is used to map it into a latent feature map of size $m_w \times m_h \times c_{\text{out}}$. Then, we apply a global average pooling layer to reduce the feature map to a generative feature vector $F_g \in \mathbb{R}^{c_{\text{out}}}$, which represents the extracted information of the input. We feed this feature vector to a classification layer, denoted as $c_g(\cdot)$, to obtain predicted probabilities. To reconstruct the input image, we use a decoder to map the latent feature map back to the original size $w \times h \times c_{\text{in}}$. The overall architecture of such a generative feature learning model is illustrated in Fig. 3.

During the training process, the CAE optimizes the model parameters in $f_g^{\text{en}}$ and $c_g$ to minimize the disparity between the predicted probabilities and the ground truth labels. This disparity is typically quantified using the cross-entropy loss

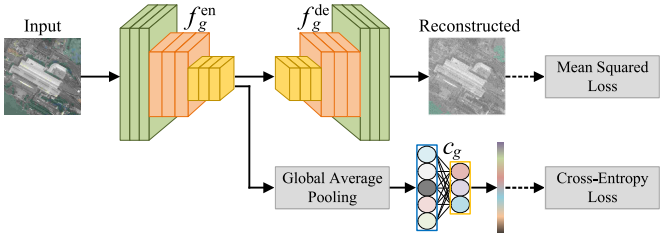



Fig. 3. Overview of the utilized CAE for generative feature learning.

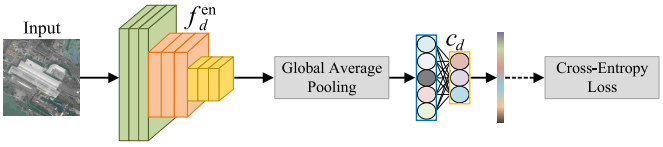



Fig. 4. Overview of the utilized CNN for discriminative feature learning.

function, expressed as follows:

$$L_{\text{ce}}\left(f_g^{\text{en}}, c_g, x_i, \boldsymbol{y}_i\right) = -\sum_{k=1}^{K} y_{ik} \log \frac{\exp\left(c_g^{(k)}(F_g(x_i))\right)}{\sum_{j=1}^{K} \exp\left(c_g^{(j)}(F_g(x_i))\right)} \tag{1}$$

where $i$ is the index of the $i$th training sample, $k$ is the index of the $k$th predicted class, and $c_j$ is the $j$th output dimension of the classification layer. The CAE also trains the model parameters in $f_g^{\text{en}}$ and $f_g^{\text{de}}$ to minimize the mean squared loss function between the original and reconstructed inputs. The mean squared loss function is defined as

$$L_{\text{mse}}\left(f_g^{\text{en}}, f_g^{\text{de}}, x_i\right) = \left(x_i - f_g^{\text{de}}\left(f_g^{\text{en}}(x_i)\right)\right)^2. \tag{2}$$

### B. Discriminative Feature Learning

We use a CNN as our discriminative feature learning model to learn discriminative features from the data. The CNN consists of multiple stacked convolution and pooling layers that form the mapping function $f_d^{\text{en}}(\cdot)$, which encodes the input to a latent representation space. Then, we apply a global average pooling layer to obtain the discriminative features $F_d$, and a classification layer parameterized by $c_d$ to obtain the predicted probabilities. Fig. 4 shows the overall architecture of our discriminative feature learning model.

Similar to (1), the loss function of discriminative feature learning can be expressed as

$$L_{\text{ce}}\left(f_d^{\text{en}}, c_d, x_i, \boldsymbol{y}_i\right) = -\sum_{k=1}^{K} y_{ik} \log \frac{\exp\left(c_d^{(k)}(F_d(x_i))\right)}{\sum_{j=1}^{K} \exp\left(c_d^{(j)}(F_d(x_i))\right)}. \tag{3}$$

### C. Trusted Joint Feature Learning Framework

In this section, we propose a joint feature learning framework that integrates generative and discriminative feature learning methods in a reliable way. The framework consists of two steps: uncertainty quantification and evidence integration. First, we modify the conventional generative and discriminative models to estimate classification uncertainty using the evidence-parameterized Dirichlet distribution. Second, we use the results of uncertainty quantification as a measure of feature

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:12:10 UTC from IEEE Xplore. Restrictions apply.

reliability and apply the Dempster–Shafer theory to combine them. This way, we can adjust the feature importance during integration and avoid unreliable local information. We explain each step in Sections III-C1–III-C3.

*1) Uncertainty Quantification:* The widely used softmax activation function is adopted to generate predictions in conventional generative and discriminative feature learning models, as shown in Figs. 3 and 4. Let $c(F(x))$ be the output logits of the model for an input $x$. Then, the softmax-based predicted probabilities for the $k$th category is given by

$$\text{Softmax}^{(k)}(c(F(x))) = \frac{\exp(c^{(k)}(F(x)))}{\sum_{j=1}^{K}\exp(c^{(j)}(F(x)))}. \tag{4}$$

The softmax function is commonly used to produce probabilities for a $K$-class classification problem. It maps each dimension of its input to the interval $[0, 1]$ and normalizes them to sum to 1. However, it also constrains the softmax output to lie on a simplex, which limits the ability to capture the uncertainty of the prediction [47]. For example, the cross-entropy loss between the softmax output and the one-hot encoded ground truth values pushes the predicted probability to be close to 0 or 1, depending on the label. This results in a point estimation that ignores the second-order statistics, especially the variance, of the predictive distribution. Consequently, some incorrect predictions may have high confidence, making it hard to quantify the uncertainty of the output.

Quantifying output uncertainty is crucial for Bayesian neural networks, which infer it indirectly from the uncertainty of network weights. However, this approach is computationally expensive. To address this issue, inspired by Sensoy et al. [43], we model uncertainty explicitly using subjective logic theory [42], which allows us to express and combine uncertainty and trust in a principled way.

Specifically, we represent the output of both generative and discriminative models as a distribution of predicted probabilities, rather than their point estimates. Let $\boldsymbol{\sigma} = [\sigma_1, \ldots, \sigma_K]$ denote the predicted probabilities for $K$ mutually exclusive categories, i.e., the $c(F(x))$ in (4), where each $\sigma_i \in [0, 1]$ and $\sum_i \sigma_i = 1$, we know that these probabilities should obey a multinomial distribution which describes the probability of $K$ mutually exclusive events. For a conventional model, the process of learning to obtain a multinomial distribution parameterized by $\boldsymbol{\sigma}$ is denoted as

$$\boldsymbol{y} \sim P(\boldsymbol{y}|\boldsymbol{\sigma}) = \text{Mult}(\boldsymbol{y}|\boldsymbol{\sigma}). \tag{5}$$

To model the variance of distribution $\text{Mult}(\boldsymbol{y}|\boldsymbol{\sigma})$, we use its conjugate prior, i.e., a Dirichlet distribution, as a substitute, which can be seen as a factory of multinomial distributions. Therefore, the output of the conventional models is transformed from the probability assignment over the possible categories to the density of each such probability assignment to capture the second-order statistics of the predicted probabilities.

Next, we induce the specific definition of this Dirichlet distribution based on subjective logic theory, which provides a well-defined theoretical framework to quantify uncertainty. Subjective logic assigns belief masses to the probabilities that reflect the degrees of confidence in different events. Thus, the model can express "*I do not know*" as an opinion for the truth over possible states. Let $b_i$ denote the belief mass for the $i$th class. The relationship between confidence degree and uncertainty can be expressed as

$$u + \sum_{i=1}^{K} b_i = 1 \tag{6}$$

where $u$ represents the overall uncertainty mass and $u, b_i \geq 0$. Belief mass $\boldsymbol{b}$ and uncertainty $u$ are computed using the evidence, noted as $\boldsymbol{e} = [e_1, \ldots, e_K]$, which are termed a measure of the amount of support collected from data in favor of a sample to be classified into a certain class. The relationship between $u$, $\boldsymbol{b}$ and $\boldsymbol{e}$ can be expressed as

$$b_i = \frac{e_i}{S}, \quad u = \frac{K}{S} \tag{7}$$

where $S = \sum_i (e_i + 1)$. It is clear that the evidence in each category is proportional to the belief mass, while the uncertainty is inversely proportional to the total evidence. For possible values of the probability mass function $\boldsymbol{\sigma}$, its probability density function that is parameterized by the evidence $\boldsymbol{e}$, i.e., the Dirichlet distribution, can be formulated as

$$Dir(\boldsymbol{\sigma}|\boldsymbol{\alpha}) = \begin{cases} \dfrac{1}{B(\boldsymbol{\alpha})}\displaystyle\prod_{i=1}^{K}\sigma_i^{\alpha_i - 1}, & \text{for } \boldsymbol{\sigma} \in \mathcal{S}_K, \\ 0, & \text{otherwise} \end{cases} \tag{8}$$

where $\alpha_i = e_i + 1$ is the parameter of the Dirichlet distribution, $B(\boldsymbol{\alpha})$ is the $K$-dimensional multinomial beta function, and $\mathcal{S}_K$ is the $K$-dimensional unit simplex

$$\mathcal{S}_K = \left\{ \boldsymbol{\sigma} \,\middle|\, \sum_{i=1}^{K} \sigma_i = 1 \text{ and } 0 \leq \sigma_1, \sigma_2, \ldots, \sigma_K \leq 1 \right\}. \tag{9}$$

Based on the aforementioned definitions, the learning objective is transformed from (5) to the following one:

$$\boldsymbol{\sigma} \sim P_\theta(\boldsymbol{\sigma}|x) = Dir(\boldsymbol{\sigma}|\boldsymbol{\alpha}) \tag{10}$$

where $\theta$ denotes the parameters of the network model. (10) means that for an input sample $x$, the conjugate prior to the conventional softmax-based learning objectives are considered. Once the parameters of $\text{Dir}(\boldsymbol{\sigma}|\boldsymbol{\alpha})$ are calculated, we can get the predicted probabilities by its mean

$$\hat{\sigma}_i = \frac{\alpha_i}{S}. \tag{11}$$

To implement the learning objective (10), we replace the softmax function with a nonnegative activation function after the classification layer in conventional models. This ensures that the evidence collected for each category is represented by a positive value. Specifically, the softplus activation function is used to obtain the evidence, which can be expressed as

$$\boldsymbol{e} = \log(1 + \exp(c(F(x)))). \tag{12}$$

Considering $\text{Dir}(\boldsymbol{\sigma}|\boldsymbol{\alpha})$ as the conjugate prior of $\text{Mult}(\boldsymbol{y}|\boldsymbol{\sigma})$, we can obtain the adjusted cross-entropy loss [43] by making a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:12:10 UTC from IEEE Xplore. Restrictions apply.

simple modification of the classical cross-entropy loss function

$$\begin{aligned} &L_{\text{ace}}(f^{\text{en}}, c, x_i, \boldsymbol{y}_i) \\ &= \int L_{\text{ce}}(f^{\text{en}}, c, x_i, \boldsymbol{y}_i) \frac{1}{B(\boldsymbol{\alpha}_i)} \prod_{j=1}^{K} \sigma_{ij}^{\alpha_{ij}-1} d\boldsymbol{\sigma}_i \\ &= \sum_{j=1}^{K} y_{ij} (\psi(S_i) - \psi(\alpha_{ij})) \end{aligned} \tag{13}$$

where $\psi(\cdot)$ is the digamma function. From the definition, it is clear that optimizing this loss function leads to a $\boldsymbol{\sigma}$ that satisfies $\sigma_i \gg \sigma_j$, where $i$ is the ground truth label and $j \neq i$. It means that (13) encourages the collection of more evidence on the correctly classified categories while expecting the misleading evidence to be as small as possible. However, (13) does not place a constraint on the samples that cannot be correctly classified. In this case, we tend to reduce all evidence to zero, where the $\text{Dir}(\boldsymbol{\sigma}|\boldsymbol{\alpha})$ corresponds to a uniform distribution, and the uncertainty is 1. This can be achieved by adding the following loss term based on Kullback-Leibler (KL)-divergence:

$$\begin{aligned} &\text{KL}[\text{Dir}(\boldsymbol{\sigma}_i|\tilde{\boldsymbol{\alpha}}_i)||\text{Dir}(\boldsymbol{\sigma}_i|[1, 1, \ldots, 1])] \\ &= \log\left(\frac{\Gamma\left(\sum_{k=1}^{K} \tilde{\alpha}_{ik}\right)}{\Gamma(K)\prod_{k=1}^{K}\Gamma\left(\tilde{\alpha}_{ik}\right)}\right) \\ &\quad + \sum_{i=1}^{K} \left(\tilde{\alpha}_{ik} - 1\right)\left[\psi\left(\tilde{\alpha}_{ik}\right) - \psi\left(\sum_{j=1}^{K} \tilde{\alpha}_{ij}\right)\right] \end{aligned} \tag{14}$$

where $\tilde{\boldsymbol{\alpha}}_i = \boldsymbol{y}_i + (\boldsymbol{1} - \boldsymbol{y}_i) \odot \boldsymbol{\alpha}_i$ and $\Gamma(\cdot)$ is the gamma function. We denote the above-mentioned loss function as $L_{\text{KL}}(f^{\text{en}}, c, x_i, \boldsymbol{y}_i)$, and the predictive loss function of the modified generative or discriminant model with the capability of uncertainty quantification can be expressed as

$$L_{\text{uc}}(f^{\text{en}}, c, x_i, \boldsymbol{y}_i) = L_{\text{ace}} + \lambda_t L_{\text{KL}} \tag{15}$$

where $\lambda_t$ is the tradeoff for constraints imposed on correctly and incorrectly classified samples. As training proceeds, $\lambda_t$ is gradually adjusted to prevent the model from excessively focusing on misclassified samples in the early stages. The optimization of (15) will result in a model that not only extracts the features but also quantifies the uncertainty of the extracted features by calculating the belief mass and uncertainty through the collected evidence.

*2) Evidence Integration:* In this section, we propose a method to integrate generative and discriminative models in a reliable way, based on the quantification of uncertainty in their outputs that we presented in Section III-C1. Our method uses the Dempster–Shafer theory, a framework for combining evidence from different sources and reducing uncertainty, leading to more reliable and accurate classification results.

Let $\mathbb{M}^v$ be the probability mass assignments from the generative and discriminative models, where $v = \{g, d\}$ and $\mathbb{M}^v = \{\{b_k^v\}_{k=1}^K, u^v\}$. We can combine them into a single probability mass assignment $\mathbb{M}$ using the Dempster–Shafer combination rule [48], as follows:

$$\mathbb{M} = \mathbb{M}^g \oplus \mathbb{M}^d. \tag{16}$$

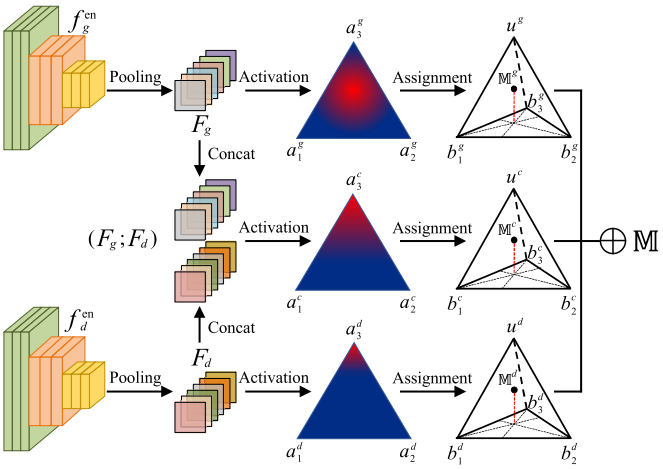



Fig. 5. Overview of the evidence integration, both feature and decision perspectives.

The detailed calculation is defined as

$$b_k = \frac{1}{1-C}\left(b_k^g b_k^d + b_k^g u^d + b_k^d u^g\right), \quad u = \frac{1}{1-C} u^g u^d \tag{17}$$

where $C = \sum_{i \neq j} b_i^g b_j^d$ is a measure of the amount of conflict between the two mass sets. It can be seen from the definition that the combination rule satisfies the following properties [44]: 1) it is commutative, i.e., $\mathbb{M}^g \oplus \mathbb{M}^d = \mathbb{M}^d \oplus \mathbb{M}^g$. 2) It preserves the level of belief, i.e., if the uncertainty of $\mathbb{M}^g$ and $\mathbb{M}^d$ are close to 1, then that of $\mathbb{M}$ is also close to 1, the belief mass $\boldsymbol{b}$ is low, and vice versa. 3) It is robust to the uncertainty of either $\mathbb{M}^g$ or $\mathbb{M}^d$, i.e., if one of the two uncertainties is high, this will not affect the combined result significantly. 4) It increases the uncertainty of the combined result when the belief masses of $\mathbb{M}^g$ and $\mathbb{M}^d$ diverge, i.e., if the conflict coefficient $C$ is large, then the uncertainty of $\mathbb{M}$ will also be large.

Based on the $\mathbb{M}$ derived from this combination rule, we can compute the corresponding evidence for each category and model the Dirichlet distribution of the combined probabilities. This way, we achieve a trusted integration of discriminative and generative models at the decision-making perspective that leverages their complementary strengths. However, the information integration from the feature perspective is also essential for a well-designed algorithm. To achieve this goal, we concatenate the features of the generative and discriminative models to collect their joint evidence, noted as $\mathbb{M}^c$. Therefore, in the proposed learning framework, there are three sets of collected evidence to be integrated and the final probability mass assignment can be calculated as

$$\mathbb{M} = (\mathbb{M}^g \oplus \mathbb{M}^d) \oplus \mathbb{M}^c \tag{18}$$

where $\mathbb{M}^g \oplus \mathbb{M}^d$ means the integration at decision-making perspective and $(\mathbb{M}^g \oplus \mathbb{M}^d) \oplus \mathbb{M}^c$ means the integration at both feature and decision-making perspectives. Fig. 5 illustrates the above-mentioned process in detail.

*3) Overall Learning Objective:* From the aforementioned description of the proposed framework for trusted generative-discriminative joint feature learning, it is clear that the framework can be categorized into four distinct parts, each

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:12:10 UTC from IEEE Xplore. Restrictions apply.

with its own learning objectives: generative feature learning, discriminative feature learning, integration of generative and discriminative features, and integration of multiple decisions. Consequently, we can derive the overall learning loss function in the following format, wherein each term corresponds to one of these four components

$$L_{\text{Total}} = L_{\text{G}} + L_{\text{D}} + L_{\text{IF}} + L_{\text{ID}}. \tag{19}$$

In the following, we analyze each loss function term in (19). $L_{\text{G}}$ is the loss function for generative feature learning, which includes both predictive loss and reconstruction loss. According to (15) and the definition of $\mathbb{M}$, its detailed formulation can be written as

$$L_{\text{G}} = \sum_{i=1}^{N} \left( L_{\text{uc}}\left(\mathbb{M}_i^g\right) + L_{\text{mse}}\left(f_g^{\text{en}}, f_g^{\text{de}}, x_i\right)\right) \tag{20}$$

where $\mathbb{M}_i^g$ denotes the probability mass assignment of the generative model to the $i$th training sample, which is correlated with the model parameters in $f_g^{\text{en}}$ and $c_g$. In $L_{\text{D}}$ and $L_{\text{IF}}$, there are only predictive losses with the variable of unintegrated probability mass assignment, and they can be expressed in the following forms:

$$L_{\text{D}} = \sum_{i=1}^{N} L_{\text{uc}}\left(\mathbb{M}_i^d\right), \quad L_{\text{IF}} = \sum_{i=1}^{N} L_{\text{uc}}\left(\mathbb{M}_i^c\right) \tag{21}$$

where the former corresponds to the learning of the discriminative model, while the latter corresponds to the learning of generative and discriminative features integration. Similar to (21), the $L_{\text{ID}}$ can be written as

$$L_{\text{ID}} = \sum_{i=1}^{N} L_{\text{uc}}(\mathbb{M}_i) = \sum_{i=1}^{N} L_{\text{uc}}\left(\mathbb{M}_i^g \oplus \mathbb{M}_i^d \oplus \mathbb{M}_i^c\right). \tag{22}$$

During training, $L_{\text{G}}$ and $L_{\text{D}}$ learn corresponding features while simultaneously estimating their uncertainty. Meanwhile, $L_{\text{IF}}$ performs feature integration while evaluating the uncertainty of integrated features. Finally, $L_{\text{ID}}$ integrates evidence from the generative, discriminative, and integrated features, combining their uncertainties to generate the final predicted probabilities. Collectively, these components ensure accurate RSI classification while addressing the unreliabilities of local information from an evidence integration perspective.

## IV. Experiments

### A. Dataset Descriptions

This section introduces the dataset utilized in our experiments. To verify the validity of our proposed framework, we conduct experiments using data obtained from three satellite RSI benchmarks: UC Merced land use dataset (UCM) [49], aerial image dataset (AID) [50], and Northwestern Polytechnical University RSI scene classification dataset (NWPU) [51]. Besides, we also validate the proposed method on a shipboard RSI dataset that we acquired ourselves [52].

*1) UCM Dataset:* This is the most important dataset for scene recognition. It contains 2100 images whose sizes are all 256 × 256 and have a pixel resolution of 0.3 m. The images equally belong to 21 categories. Thus, each category has 100 images. All samples are manually extracted from the large images in the United States Geological Survey National Map Urban Area Imagery Database collected from various urban areas around the country. Two commonly used evaluation protocols are 50%/50% and 80%/20% for training/testing. Some examples in this dataset are shown in Fig. 6.

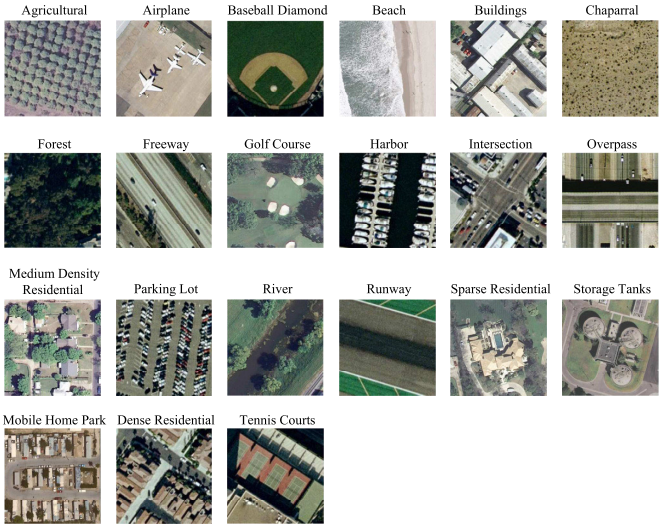



Fig. 6. Samples of the UCM dataset.




Fig. 7. Samples of the AID dataset.

*2) AID Dataset:* This is a challenging dataset that is generated by collecting the images from multisource sensors on Google Earth. It has high intraclass diversities since the images are carefully chosen from different countries. They are extracted at different times and seasons under different imaging conditions. It has 10 000 images at the size of 600 × 600 belonging to 30 categories, where each category has 220 to 420 samples. Two evaluation protocols are 20%/80% and 50%/50% for training/testing. Some examples in this dataset are shown in Fig. 7.

*3) NWPU Dataset:* This dataset is characterized by a great number of samples. It contains 31 500 images and 45 categories in total, where each category has 700 samples. Each image has 256 × 256 pixels. The spatial resolutions are varied

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:12:10 UTC from IEEE Xplore. Restrictions apply.

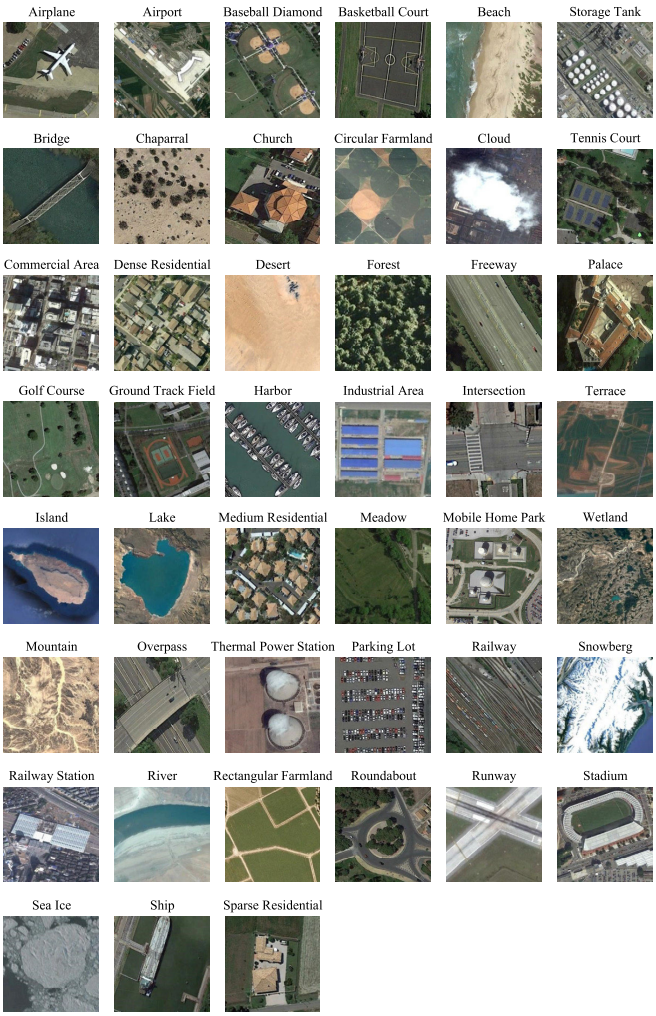



Fig. 8. Samples of the NWPU dataset.

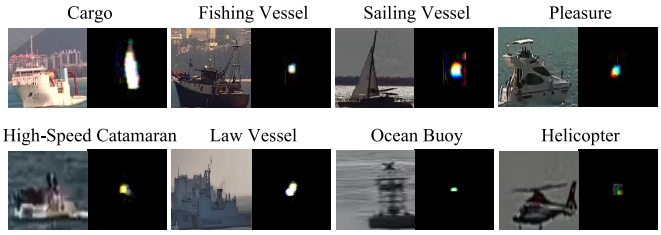



Fig. 9. Samples of the shipboard RSI dataset.

from 0.2 to 30 m. Some special landforms, such as islands, lakes, regular mountains, and snow mountains, maybe in lower resolutions. Two evaluation protocols are 10%/90% and 20%/80% for training/testing. Some examples in this dataset are shown in Fig. 8.

*4) Shipboard RSI Dataset:* This dataset is collected through an offshore workstation we established in Sanya, China, equipped with optical and radar sensors. The optical images are collected from a Hikvision camera and the radar plan-position indicator images are collected by Furuno FAR-2117. A large number of sea trials have been conducted to collect sufficient data. It contains more than 26 000 images at the size of 224 × 224 belonging to eight categories. Some examples in this dataset are shown in Fig. 9.

### B. Experimental Settings

We conduct our experiments on four NVIDIA RTX TITAN GPUs. To implement the related algorithms, we use the PyTorch deep learning platform. In the following, we provide details on the implementation of our experiments.

In our experiments on three RSI benchmark datasets, we follow the experimental settings reported in [53]: When using the CNN as the backbone, we use two MillionAID dataset [54] pretrained ResNet50 models as the encoder of generative model and the backbone of discriminative model. When using the vision transformer (ViT) as a backbone, the MillionAID pretrained Swin-transformer-tiny (Swin-T) model is selected as an alternative. The decoder of the generative model is the transposed convolution version of the encoder, but it is randomly initialized. These models are fine-tuning 200 epochs with a batch size of 64 under the proposed trusted joint feature learning framework. During this process, the AdamW optimizer is adopted where the momentum is set to 0.9 and the weight decay is set to 0.05. The initial learning rate is 1e-3, which is adjusted by the cosine scheduling policy with a minimal learning rate of 5e-6. In addition, we set the warming-up epochs to 5, where the learning rate is set to 5e-7. During training, All the input image is resized to 224 × 224 by randomly cropping during training. During testing, the image at the same size is obtained through a center crop. In addition, a series of data augmentations, including AutoAugment, random erasing, Mixup, CutMix, and color jitter, are applied to improve the training performance.

In our experiments on the shipboard RSI dataset, the models are randomly initialized and a stochastic gradient descent optimizer with the momentum of 0.9 is used to minimize the objective functions. During training, the involved models are trained for 300 epochs with an initial learning rate of 0.1. The weight decay is set as 5e-4. The learning rate decreases to 1e-6 at the end of training.

To evaluate the effectiveness of our proposed feature learning framework, several recently proposed generative-based and discriminative-based RSI classifiers are used as comparisons. These comparing methods either employ CNN or ViT as the feature extraction backbone. Most of them incorporate multibranch or multiscale operations to achieve the fusion of multiple types of features, which is closely related to the theme of our proposed method. The overall accuracy is chosen as the evaluation metric with the format of "average ± deviation" based on ten independent runs.

### C. Experimental Results

Tables I and II illustrate the classification performance and computational complexity comparisons of CNN-based and ViT-based state-of-the-art approaches with the proposed method, respectively. Among them, FLOPs represent floating point operations, used to evaluate computational complexity, while "remote sensing pretraining (RSP)" stands for utilizing backbone models pretrained on RSIs [53]. Tables III and V show the results of our proposed method and other comparing methods, including multiple generative models and classical discriminative models, on three RSI benchmark datasets.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:12:10 UTC from IEEE Xplore. Restrictions apply.

TABLE I

TESTING ACCURACY (%) AND COMPUTATIONAL COMPLEXITY OF OUR PROPOSED METHOD AND OTHER CNN-BASED STATE-OF-THE-ART METHODS ON THREE REMOTE SENSING SCENE CLASSIFICATION DATASETS. THE TOP TWO RESULTS ARE HIGHLIGHTED IN RED AND BLUE. † INDICATES THAT THERE ARE TWO BACKBONE BRANCHES, AND ‡ INDICATES THAT THERE ARE SINGLE BACKBONES WITH MULTISCALE OPERATIONS

| Method | UCM (50%) | UCM (80%) | AID (20%) | AID (50%) | NWPU (10%) | NWPU (20%) | FLOPs |
|---|---|---|---|---|---|---|---|
| D-CNNs [34] | - | 98.93±0.10 | 90.82±0.16 | 96.89±0.10 | 89.22±0.50 | 91.89±0.22 | 15.5G |
| CNN-CapsNet‡ [55] | 97.59±0.16 | 99.05±0.24 | 93.79±0.13 | 96.32±0.12 | 89.03±0.21 | 92.60±0.11 | 3.6G |
| SCCov‡ [56] | - | 99.05±0.25 | 93.12±0.25 | 96.10±0.16 | 89.30±0.35 | 92.10±0.25 | 15.5G |
| SKAL† [57] | 98.19±0.05 | 99.70±0.30 | 94.26±0.15 | 96.65±0.11 | 90.41±0.12 | 92.95±0.09 | 3.0G |
| MBLANet‡ [58] | - | 99.64±0.12 | 95.60±0.17 | 97.14±0.03 | 92.32±0.15 | 94.66±0.11 | 4.5G |
| MGML-FENet† [59] | - | 99.81±0.10 | 94.47±0.15 | 97.89±0.07 | 90.69±0.14 | 93.36±0.12 | 36.5G |
| LSENet‡ [60] | 98.53±0.37 | 99.78±0.18 | 94.41±0.16 | 96.36±0.19 | 92.23±0.14 | 93.34±0.15 | 15.5G |
| MLFFRGKA† [61] | - | - | 95.20±0.17 | 97.26±0.16 | 91.02±0.11 | 93.59±0.07 | 3.6G |
| MS2AP‡ [62] | 98.38±0.35 | 99.01±0.42 | 95.42±0.28 | 96.86±0.20 | 92.27±0.21 | 93.91±0.15 | 11.6G |
| MBFANet‡ [63] | - | 99.66±0.19 | 93.98±0.15 | 96.93±0.16 | 91.61±0.14 | 94.01±0.08 | 4.5G |
| ResNet50 RSP [53] | - | 99.48±0.10 | 96.81±0.03 | 97.89±0.08 | 93.93±0.10 | 95.02±0.06 | 4.1G |
| EAM [64] | - | 99.21±0.26 | 94.26±0.11 | 97.06±0.19 | 91.91±0.22 | 94.29±0.09 | 7.9G |
| **Ours**† | 98.62±0.11 | 99.71±0.09 | 96.53±0.12 | 98.02±0.16 | 94.25±0.15 | 95.66±0.13 | 8.3G |
| **Ours RSP**† | 98.93±0.09 | 99.89±0.13 | 96.92±0.10 | 98.51±0.09 | 94.32±0.12 | 95.97±0.17 | 8.3G |

TABLE II

TESTING ACCURACY (%) AND COMPUTATIONAL COMPLEXITY OF OUR PROPOSED METHOD AND OTHER VIT-BASED STATE-OF-THE-ART METHODS ON THREE REMOTE SENSING SCENE CLASSIFICATION DATASETS

| Method | UCM (50%) | UCM (80%) | AID (20%) | AID (50%) | NWPU (10%) | NWPU (20%) | FLOPs |
|---|---|---|---|---|---|---|---|
| ViT‡ [65] | 98.48±0.39 | 99.01±0.21 | 94.90±0.29 | 96.49±0.18 | 91.59±0.19 | 93.90±0.20 | 16.9G |
| CTNet† [66] | - | - | 96.25±0.10 | 97.70±0.11 | 93.90±0.14 | 95.40±0.15 | 20.5G |
| TRS‡ [67] | 98.76±0.13 | 99.52±0.17 | 95.54±0.18 | 98.48±0.06 | 93.06±0.11 | 95.56±0.20 | 8.4G |
| MITformer‡ [68] | 98.91±0.27 | 99.83±0.24 | 96.04±0.21 | 97.96±0.18 | 94.09±0.15 | 95.93±0.17 | 17.8G |
| LG-ViT† [69] | 99.35±0.16 | 99.93±0.12 | 96.55±0.21 | 97.67±0.15 | 93.87±0.22 | 95.60±0.20 | 5.4G |
| Swin-T‡ [70] | - | 99.62±0.19 | 96.55±0.03 | 98.10±0.06 | 92.73±0.09 | 94.70±0.10 | 4.4G |
| Swin-T RSP‡ [53] | - | 99.52±0.00 | 96.83±0.08 | 98.30±0.04 | 93.02±0.12 | 94.51±0.05 | 4.4G |
| **Ours**† | 99.37±0.09 | 99.81±0.12 | 96.89±0.07 | 98.49±0.07 | 94.06±0.09 | 96.11±0.04 | 8.7G |
| **Ours RSP**† | 99.23±0.08 | 99.86±0.11 | 97.03±0.05 | 98.58±0.11 | 94.13±0.12 | 96.03±0.17 | 8.7G |

Fig. 10 demonstrates the ablation performance of the proposed method on the shipboard RSI dataset, which reflects the impact of different backbones, backbone ensemble, and evidential fusion on the classification accuracy. Based on these results, we offer the following analyses.

*1) Results on UCM Dataset:* To demonstrate the effectiveness of our proposed framework, we conduct experimental comparisons of the involved RSI classifiers on the UCM dataset. Results are presented as percentages in the leftmost two columns of Tables I and II, as well as in Table III.

By analyzing the quantitative experimental results, we can draw the following conclusions: although generative models, including DAEs- and GANs-based approaches, extract features that exhibit significant differences from discriminative models (as shown in Fig. 1), there still exists a certain gap in classification accuracy between them. Nevertheless, the best-performing PS-GANs in generative models have comparable accuracy to ResNet50, highlighting their potential to enhance the performance of discriminative models. We confirm this by comparing our proposed method with various discriminative models, including classic CNNs and recent ViTs. By incorporating the information from generative models in a trusted manner, the proposed method stands out among numerous models with CNN as the backbone, achieving state-of-the-art performance in both training scenarios. Even without considering the use of RSP, the proposed method attains the highest accuracy with 50% of training samples, and when 80% samples are used for training, it lags behind the best-performing model in this scenario, MGML-FENet, by only 0.1%, while significantly outperforming in terms of FLOPs. Similar conclusions can also be drawn from the experimental results of models with ViT as the backbone,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:12:10 UTC from IEEE Xplore. Restrictions apply.

TABLE III
TESTING ACCURACY (%) OF GENERATIVE, CLASSICAL DISCRIMINATIVE, AND OUR PROPOSED METHODS ON UCM DATASET

| Type | Method | UCM (50%) | UCM (80%) |
|---|---|---|---|
| Generative | HCSAE [26] | - | 97.14±1.19 |
| | SCDAE [71] | - | 93.70±1.30 |
| | SGUFL [23] | - | 82.72±1.18 |
| | MARTA GANs [72] | 85.50±0.69 | 94.86±0.82 |
| | MF-WGANs [73] | - | 98.40±0.32 |
| | Attention GANs [74] | 89.06±0.50 | 97.69±0.69 |
| | EM-GANs [75] | 91.93±0.29 | 95.68±0.42 |
| | PS-GANs [76] | 96.55±0.42 | 98.48±0.64 |
| Discriminative | GoogLeNet [50] | 92.70±0.60 | 94.31±0.89 |
| | VGG16 [50] | 94.14±0.69 | 95.21±1.20 |
| | ResNet50 [77] | - | 98.69±0.49 |
| | $\text{Ours}_{\text{res-50}}$ | 98.62±0.11 | 99.71±0.09 |
| | $\text{Ours}_{\text{res-50}}$ RSP | 98.93±0.09 | **99.89±0.13** |
| | $\text{Ours}_{\text{swin-t}}$ | **99.37±0.09** | 99.81±0.12 |
| | $\text{Ours}_{\text{swin-t}}$ RSP | 99.23±0.08 | 99.86±0.11 |

TABLE IV
TESTING ACCURACY (%) OF GENERATIVE, CLASSICAL DISCRIMINATIVE, AND OUR PROPOSED METHODS ON AID DATASET

| Type | Method | AID (20%) | AID (50%) |
|---|---|---|---|
| Generative | MARTA GANs [72] | 75.39±0.49 | 81.57±0.33 |
| | MF-WGANs [73] | - | 80.35±1.54 |
| | SGSAGGANs [78] | 81.66±0.33 | 87.77±0.36 |
| | Attention GANs [74] | 78.95±0.23 | 84.52±0.18 |
| | EM-GANs [75] | - | 97.68±0.25 |
| | PS-GANs [76] | 90.27±0.55 | 93.47±0.70 |
| Discriminative | GoogLeNet [50] | 83.44±0.40 | 86.39±0.55 |
| | VGG16 [50] | 86.59±0.29 | 89.64±0.36 |
| | ResNet50 [77] | 93.64±0.25 | 96.62±0.13 |
| | $\text{Ours}_{\text{res-50}}$ | 96.53±0.12 | 98.02±0.16 |
| | $\text{Ours}_{\text{res-50}}$ RSP | 96.92±0.10 | 98.51±0.09 |
| | $\text{Ours}_{\text{swin-t}}$ | 96.89±0.07 | 98.49±0.07 |
| | $\text{Ours}_{\text{swin-t}}$ RSP | **97.03±0.05** | **98.58±0.11** |

where the proposed method demonstrates comparable classification performance and computational complexity to LG-ViT, indicating the validity of trusted generative-discriminative joint feature learning.

*2) Results on AID Dataset:* The quantitative results of the comparative experiments on the AID dataset are shown in the middle two columns of Tables I and II, as well as in Table IV.

The experimental results of this subsection exhibit a slight deviation from the trends observed in the previous results: with the increased difficulty stemming from the 30-class classification task, the performance improvement introduced by the proposed method becomes more significant. First, compared with classical generative and discriminative models under equivalent conditions, the proposed method achieves a significant accuracy improvement of approximately 3% (20% samples training) and 2.4% (50% samples training). Then,

TABLE V
TESTING ACCURACY (%) OF GENERATIVE, CLASSICAL DISCRIMINATIVE, AND OUR PROPOSED METHODS ON NWPU DATASET

| Type | Method | NWPU (10%) | NWPU (20%) |
|---|---|---|---|
| Generative | MARTA GANs [72] | 68.63±0.22 | 75.03±0.28 |
| | MF-WGANs [73] | - | 79.49±1.27 |
| | SGSAGGANs [78] | 75.03±0.52 | 81.75±0.23 |
| | Attention GANs [74] | 72.21±0.21 | 77.99±0.19 |
| | PS-GANs [76] | 86.50±0.62 | 89.63±0.93 |
| Discriminative | GoogLeNet [51] | 79.19±0.38 | 78.48±0.26 |
| | VGG16 [51] | 76.47±0.18 | 79.79±0.15 |
| | ResNet50 [77] | 88.48±0.21 | 91.86±0.19 |
| | $\text{Ours}_{\text{res-50}}$ | 94.25±0.15 | 95.66±0.13 |
| | $\text{Ours}_{\text{res-50}}$ RSP | **94.32±0.12** | 95.97±0.17 |
| | $\text{Ours}_{\text{swin-t}}$ | 94.06±0.09 | **96.11±0.04** |
| | $\text{Ours}_{\text{swin-t}}$ RSP | 94.13±0.12 | 96.03±0.17 |

in comparison with CNN-based state-of-the-art methods, the proposed method consistently achieves superior classification performance under equivalent conditions, regardless of the adoption of RSP. Particularly noteworthy is that among various models not utilizing RSP, the proposed method attains a minimum accuracy improvement of approximately 1% when trained with 20% of the samples. Finally, for ViT-based state-of-the-art methods, the two versions of the proposed method achieve optimal and suboptimal results, respectively, strongly demonstrating the positive role of generative features in improving the performance of the discriminative task.

We can use the ResNet50 and Swin-T as baselines of our proposed method since they have similar network architectures. The baseline is a discriminative model trained separately, while our method is a trusted generative-discriminative joint learning version. Comparing their performance and computational complexity, it is evident that the introduction of the proposed method, despite adding an additional network branch that doubles the computational complexity, effectively enhances the accuracy of baselines. Given the current development in computational capabilities, such a level of computational overhead is acceptable, particularly if we choose a backbone model with relatively lower FLOPs.

*3) Results on NWPU Dataset:* The rightmost two columns of Tables I and II and Table V present experimental results of the involved methods on three RSI benchmark datasets.

Analyzing the experimental results in this section leads to conclusions similar to those obtained from the AID dataset: The experimental results of the generative approaches show that the features obtained by generative learning are discriminative to some extent. Therefore, after utilizing the trusted generative-discriminative integration framework to incorporate the valid generative information, the proposed method achieves accuracy improvements over the involved state-of-the-art methods. When using 10% samples for training, the proposed method outperforms the better performing models in the CNN-based methods, including multiscale staking attention pooling (MS2AP), location and seasonality enhanced network (LSENet), multibranch local attention network (MBLANet), EAM, and MBFANet, in terms of accuracy by 1.98%, 2.02%,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:12:10 UTC from IEEE Xplore. Restrictions apply.

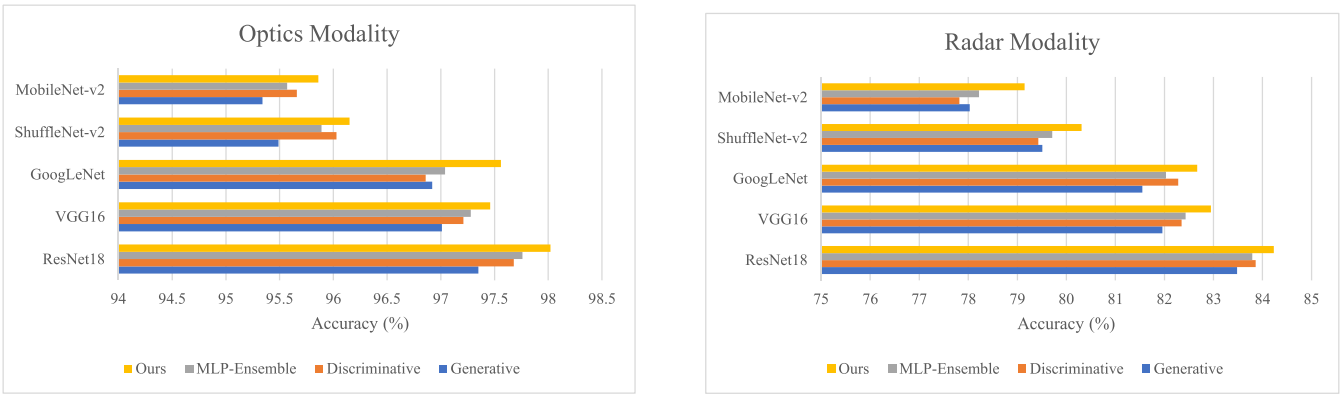



Fig. 10. Results of comparative experiments on the shipboard RSI dataset. (a) Experimental results on optics modality. (b) Experimental results on radar modality.

1.93%, 2.34%, and 2.64%, respectively, without utilizing RSP. When the training sample ratio is increased to 20%, the proposed method achieves an accuracy improvement of 1.75%, 2.32%, 1%, 1.37%, and 1.65% compared with the mentioned models. In addition, the introduction of RSP further extends the lead of the proposed method in accuracy. This establishes the proposed method as achieving both optimal and suboptimal performance in comparison with the involved CNN-based methods. For ViT-based approaches, the proposed method continues its good performance and achieves quite competitive accuracies in both training scenarios. It surpasses all other ViT-based comparing methods, with the exception of a 0.03% lower accuracy compared with MITformer when trained with 10% of the samples. However, the FLOPs of the proposed method are only about half of the MITformer model, highlighting a well-balanced tradeoff between classification performance and computational complexity. In summary, the results of these comparative experiments further demonstrate the effectiveness of our proposed trusted joint feature learning framework for improving the existing discriminative-based RSI classification methods.

*4) Results on Shipboard RSI Dataset:* In the numerical experiments of Section IV-C3, we verify the effectiveness of the proposed generative-discriminative joint learning method using three RSI benchmark datasets acquired through aerial platforms. In this part, we conduct ablation studies using an RSI dataset obtained from a maritime platform. Our aim is to test the effect of different backbones, backbones ensemble, and evidential fusion on the classification performance. In order to achieve this goal, we select five widely used architectures, including MobileNet-v2 [79], ShuffleNet-v2 [80], GoogLeNet, VGG16, and ResNet18. For each modality, we conduct separate tests for the classification performance when using each architecture individually for generative or discriminative learning. Additionally, we evaluate the performance of generative-discriminative joint learning using multilayer perceptron (MLP-ensemble) and explore the proposed evidential fusion approach for generative-discriminative joint feature learning. The experimental results are reported in Fig. 10.

The experimental results show that the discriminative models generally achieve better classification performance than the generative models, except for a few cases. Besides, a suitable method is important for the integration of generative and discriminative features. From the experimental results, it is evident that when the difference in classification accuracy between the two types of features is relatively small, directly employing an MLP for feature integration contributes to an improvement in classification accuracy in most scenarios. However, when the difference between the two sets is substantial, the situation takes a different turn. In contrast, the proposed framework enhances the performance of both generative and discriminative models in all cases by integrating them in a trusted manner, highlighting the assurance in feature integration effectiveness brought about by evidential fusion. In particular, we observe that when the generative model has a higher accuracy than the discriminative model, their combination leads to a larger improvement in accuracy.

These results further validate the effectiveness of the proposed generative-discriminative joint feature learning framework for RSI classification from the following aspects: first, the proposed method is not tailored to a specific backbone model but exhibits adaptability across various backbones. Second, the choice of a simplistic feature integration method may pose the risk of decreasing classification performance instead of enhancing it. Finally, the evidential fusion of generative and discriminative features within the proposed method plays a crucial role in improving classification performance. By incorporating evidential learning into the information integration process, the framework enables the baseline method to extract more discriminative features, resulting in significant improvements in classification accuracy. This conclusion is validated on several RSI datasets and has great potential to improve the performance of existing RSI classifiers due to its plug-and-play nature.

### D. Visualization

In this section, we apply the AID dataset to compare the attention distribution of generative and discriminative models under separate and joint training scenarios. Following the motivating experiment shown in Fig. 1(c), we use the CAM to measure the attention distribution. Fig. 11 shows the visualization results.

We select four representative categories for visualization, i.e., airport, bridge, storage tanks, and port. These categories share the characteristic of having many discriminative local

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:12:10 UTC from IEEE Xplore. Restrictions apply.

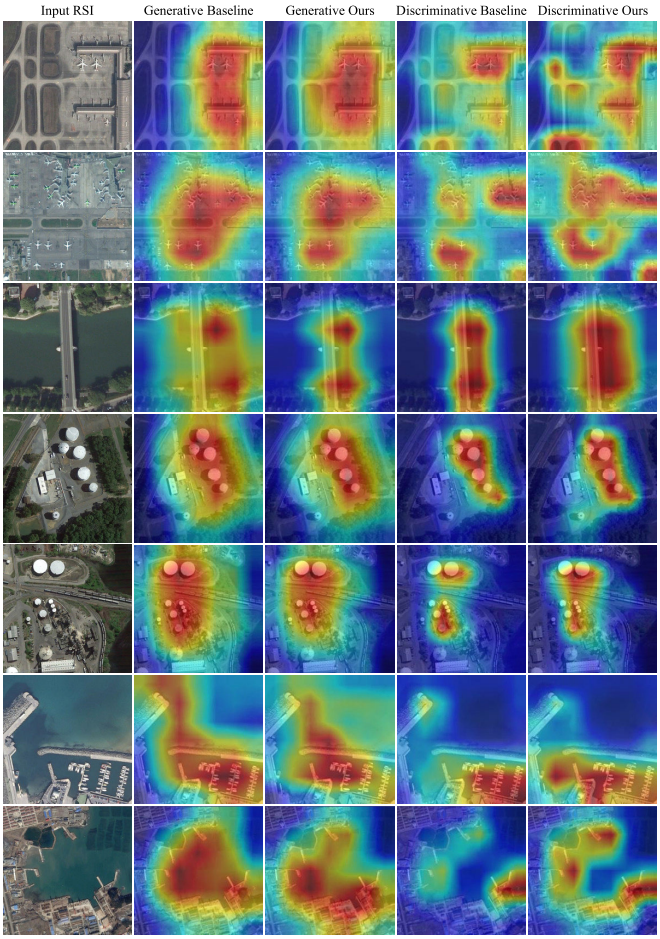



Fig. 11. Comparison of visualization results.

features in the images. Fig. 11 displays, from left to right, the input RSI, the CAM of the generative model trained separately (generative baseline), the CAM of the generative model trained jointly (generative ours), the CAM of the discriminative model trained separately (discriminative baseline), and the CAM of the discriminative model trained jointly (discriminative ours). We can clearly observe that joint training reduces the attention of the generative model to local details and increases the attention of the discriminative model to local details, reflecting the integration of two distinct types of features.

## V. Conclusion

This article proposes a novel generative-discriminative joint feature learning framework for RSI classification. It is the first work that leverages generative models to enhance the discriminative performance of RSI classifiers. The key idea is to use evidential learning to capture the uncertainty of the predictions from both generative and discriminative models, and then integrate them at the evidence level. This way, our proposed joint feature learning framework can filter out the nondiscriminative local features that are common in RSIs and focus on the most effective ones for classification. We conduct experiments on three widely used benchmark datasets and one shipboard RSI dataset and show that our framework outperforms the baseline methods.

In future work, we intend to apply the evidential learning-based trusted integration to other RSI tasks, such as multiscale, multispectral, and multisensor integration.

## References

[1] M. Datcu et al., "Information mining in remote sensing image archives: System concepts," *IEEE Trans. Geosci. Remote Sens.*, vol. 41, no. 12, pp. 2923–2936, Dec. 2003.

[2] Y. Li, Y. Zhang, X. Huang, H. Zhu, and J. Ma, "Large-scale remote sensing image retrieval by deep hashing neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 950–965, Feb. 2018.

[3] X. Lu, B. Wang, X. Zheng, and X. Li, "Exploring models and data for remote sensing image caption generation," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2183–2195, Apr. 2018.

[4] D. Lu and Q. Weng, "A survey of image classification methods and techniques for improving classification performance," *Int. J. Remote Sens.*, vol. 28, no. 5, pp. 823–870, Mar. 2007, doi: 10.1080/01431160600746456.

[5] Y. Yang and S. Newsam, "Bag-of-visual-words and spatial extensions for land-use classification," in *Proc. 18th SIGSPATIAL Int. Conf. Adv. Geographic Inf. Syst.*, Nov. 2010, pp. 270–279.

[6] R. Zhang, S. Yang, Q. Zhang, L. Xu, Y. He, and F. Zhang, "Graph-based few-shot learning with transformed feature propagation and optimal class allocation," *Neurocomputing*, vol. 470, pp. 247–256, Jan. 2022.

[7] R. Zhang et al., "Differential feature awareness network within antagonistic learning for infrared-visible object detection," *IEEE Trans. Circuits Syst. Video Technol.*, early access, Jun. 26, 2023, doi: 10.1109/TCSVT.2023.3289142.

[8] G. Cheng, X. Xie, J. Han, L. Guo, and G.-S. Xia, "Remote sensing image scene classification meets deep learning: Challenges, methods, benchmarks, and opportunities," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3735–3756, 2020.

[9] G. Li, H. Zheng, C. Wang, C. Li, C. Zheng, and D. Tao, "3DDesigner: Towards photorealistic 3D object generation and editing with text-guided diffusion models," 2022, *arXiv:2211.14108*.

[10] W. Li et al., "Stacked autoencoder-based deep learning for remote-sensing image classification: A case study of African land-cover mapping," *Int. J. Remote Sens.*, vol. 37, no. 23, pp. 5632–5646, Dec. 2016, doi: 10.1080/01431161.2016.1246775.

[11] E. Maggiori, Y. Tarabalka, G. Charpiat, and P. Alliez, "Convolutional neural networks for large-scale remote-sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 645–657, Feb. 2017.

[12] L. Wilkinson and M. Friendly, "The history of the cluster heat map," *Amer. Statistician*, vol. 63, no. 2, pp. 179–184, May 2009.

[13] Z. Xie, Z. Geng, J. Hu, Z. Zhang, H. Hu, and Y. Cao, "Revealing the dark secrets of masked image modeling," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 14475–14485.

[14] B. Zhou, A. Khosla, A. Lapedriza, A. Oliva, and A. Torralba, "Learning deep features for discriminative localization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2921–2929.

[15] S. Lei, Z. Shi, and Z. Zou, "Super-resolution for remote sensing images via local–global combined network," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 8, pp. 1243–1247, Aug. 2017.

[16] H. Zhang, Y. Liao, H. Yang, G. Yang, and L. Zhang, "A local–global dual-stream network for building extraction from very-high-resolution remote sensing images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 3, pp. 1269–1283, Mar. 2022.

[17] G. E. Hinton and R. R. Salakhutdinov, "Reducing the dimensionality of data with neural networks," *Science*, vol. 313, no. 5786, pp. 504–507, Jul. 2006.

[18] I. Goodfellow et al., "Generative adversarial networks," *Commun. ACM*, vol. 63, no. 11, pp. 139–144, 2020.

[19] G. Cheng, J. Han, L. Guo, Z. Liu, S. Bu, and J. Ren, "Effective and efficient midlevel visual elements-oriented land-use classification using VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 8, pp. 4238–4249, Aug. 2015.

[20] S. Malek, F. Melgani, Y. Bazi, and N. Alajlan, "Reconstructing cloud-contaminated multispectral images with contextualized autoencoder neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2270–2282, Apr. 2018.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:12:10 UTC from IEEE Xplore. Restrictions apply.

[21] C. Tao, H. Pan, Y. Li, and Z. Zou, "Unsupervised spectral–spatial feature learning with stacked sparse autoencoder for hyperspectral imagery classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 12, pp. 2438–2442, Dec. 2015.

[22] G. Dong, G. Liao, H. Liu, and G. Kuang, "A review of the autoencoder and its variants: A comparative perspective from target recognition in synthetic-aperture radar images," *IEEE Geosci. Remote Sens. Mag.*, vol. 6, no. 3, pp. 44–68, Sep. 2018.

[23] F. Zhang, B. Du, and L. Zhang, "Saliency-guided unsupervised feature learning for scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 4, pp. 2175–2184, Apr. 2015.

[24] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.

[25] J. Geng, J. Fan, H. Wang, X. Ma, B. Li, and F. Chen, "High-resolution SAR image classification via deep convolutional autoencoders," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 11, pp. 2351–2355, Nov. 2015.

[26] X. Han, Y. Zhong, B. Zhao, and L. Zhang, "Scene classification based on a hierarchical convolutional sparse auto-encoder for high spatial resolution imagery," *Int. J. Remote Sens.*, vol. 38, no. 2, pp. 514–536, Jan. 2017.

[27] X. Zhang, G. Chen, W. Wang, Q. Wang, and F. Dai, "Object-based land-cover supervised classification for very-high-resolution UAV images using stacked denoising autoencoders," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 7, pp. 3373–3385, Jul. 2017.

[28] N. Lv, C. Chen, T. Qiu, and A. K. Sangaiah, "Deep learning and superpixel feature extraction based on contractive autoencoder for change detection in SAR images," *IEEE Trans. Ind. Informat.*, vol. 14, no. 12, pp. 5530–5538, Dec. 2018.

[29] X. Shen, B. Liu, Y. Zhou, J. Zhao, and M. Liu, "Remote sensing image captioning via variational autoencoder and reinforcement learning," *Knowl.-Based Syst.*, vol. 203, Sep. 2020, Art. no. 105920.

[30] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," *Commun. ACM*, vol. 60, no. 6, pp. 84–90, May 2017.

[31] O. A. B. Penatti, K. Nogueira, and J. A. dos Santos, "Do deep features generalize from everyday objects to remote sensing and aerial scenes domains?" in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2015, pp. 44–51.

[32] D. Marmanis, M. Datcu, T. Esch, and U. Stilla, "Deep learning earth observation classification using ImageNet pretrained networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 1, pp. 105–109, Jan. 2016.

[33] M. Castelluccio, G. Poggi, C. Sansone, and L. Verdoliva, "Land use classification in remote sensing images by convolutional neural networks," 2015, *arXiv:1508.00092*.

[34] G. Cheng, C. Yang, X. Yao, L. Guo, and J. Han, "When deep learning meets metric learning: Remote sensing image scene classification via learning discriminative CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 5, pp. 2811–2821, May 2018.

[35] Q. Wang, S. Liu, J. Chanussot, and X. Li, "Scene classification with recurrent attention of VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 1155–1167, Feb. 2019.

[36] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[37] S. Chen, H. Wang, F. Xu, and Y.-Q. Jin, "Target classification using the deep convolutional networks for SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4806–4817, Aug. 2016.

[38] R. M. Neal, *Bayesian Learning for Neural Networks*, vol. 118. Cham, Switzerland: Springer, 2012.

[39] Y. Gal and Z. Ghahramani, "Dropout as a Bayesian approximation: Representing model uncertainty in deep learning," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2016, pp. 1050–1059.

[40] B. Lakshminarayanan, A. Pritzel, and C. Blundell, "Simple and scalable predictive uncertainty estimation using deep ensembles," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, vol. 30, 2017, pp. 6402–6413.

[41] B. Charpentier, D. Zügner, and S. Günnemann, "Posterior network: Uncertainty estimation without OOD samples via density-based pseudo-counts," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 1356–1367.

[42] A. Jøsang, *Subjective Logic: A Formalism for Reasoning Under Uncertainty*. Cham, Switzerland: Springer, 2018.

[43] M. Sensoy, L. Kaplan, and M. Kandemir, "Evidential deep learning to quantify classification uncertainty," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 31, 2018, pp. 3179–3189.

[44] Z. Han, C. Zhang, H. Fu, and J. T. Zhou, "Trusted multi-view classification with dynamic evidential fusion," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 2, pp. 2551–2566, Feb. 2023.

[45] A. P. Dempster, "Upper and lower probabilities induced by a multivalued mapping," *Ann. Math. Statist.*, vol. 38, no. 2, pp. 325–339, Apr. 1967.

[46] W. Liu, X. Yue, Y. Chen, and T. Denoeux, "Trusted multi-view deep learning with opinion aggregation," in *Proc. AAAI Conf. Artif. Intell.*, 2022, vol. 36, no. 7, pp. 7585–7593.

[47] J. Van Amersfoort, L. Smith, Y. W. Teh, and Y. Gal, "Uncertainty estimation using a single deep deterministic neural network," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2020, pp. 9690–9700.

[48] G. Shafer, *A Mathematical Theory of Evidence*. Princeton, NJ, USA: Princeton Univ. Press, 1976.

[49] Q. Zhu, Y. Zhong, B. Zhao, G.-S. Xia, and L. Zhang, "Bag-of-visual-words scene classifier with local and global features for high spatial resolution remote sensing imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 6, pp. 747–751, Jun. 2016.

[50] G.-S. Xia et al., "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3965–3981, Jul. 2017.

[51] G. Cheng, J. Han, and X. Lu, "Remote sensing image scene classification: Benchmark and state of the art," *Proc. IEEE*, vol. 105, no. 10, pp. 1865–1883, Oct. 2017.

[52] L. Si et al., "A dimensional structure based knowledge distillation method for cross-modal learning," 2023, *arXiv:2306.15977*.

[53] D. Wang, J. Zhang, B. Du, G.-S. Xia, and D. Tao, "An empirical study of remote sensing pretraining," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5608020.

[54] Y. Long et al., "On creating benchmark dataset for aerial image interpretation: Reviews, guidances, and million-AID," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4205–4230, 2021.

[55] W. Zhang, P. Tang, and L. Zhao, "Remote sensing image scene classification using CNN-CapsNet," *Remote Sens.*, vol. 11, no. 5, p. 494, Feb. 2019.

[56] N. He, L. Fang, S. Li, J. Plaza, and A. Plaza, "Skip-connected covariance network for remote sensing scene classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 5, pp. 1461–1474, May 2020.

[57] Q. Wang, W. Huang, Z. Xiong, and X. Li, "Looking closer at the scene: Multiscale representation learning for remote sensing image scene classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 4, pp. 1414–1428, Apr. 2022.

[58] S.-B. Chen, Q.-S. Wei, W.-Z. Wang, J. Tang, B. Luo, and Z.-Y. Wang, "Remote sensing scene classification via multi-branch local attention network," *IEEE Trans. Image Process.*, vol. 31, pp. 99–109, 2022.

[59] Q. Zhao, S. Lyu, Y. Li, Y. Ma, and L. Chen, "MGML: Multigranularity multilevel feature ensemble network for remote sensing scene classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 5, pp. 2308–2322, May 2023.

[60] Q. Bi, K. Qin, H. Zhang, and G.-S. Xia, "Local semantic enhanced ConvNet for aerial scene recognition," *IEEE Trans. Image Process.*, vol. 30, pp. 6498–6511, 2021.

[61] F. Li and J. Wang, "Remote sensing image scene classification via regional growth-based key area fine location and multilayer feature fusion," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[62] Q. Bi, H. Zhang, and K. Qin, "Multi-scale stacking attention pooling for remote sensing scene classification," *Neurocomputing*, vol. 436, pp. 147–161, May 2021.

[63] J. Shi, W. Liu, H. Shan, E. Li, X. Li, and L. Zhang, "Remote sensing scene classification based on multibranch fusion attention network," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[64] Z. Zhao, J. Li, Z. Luo, J. Li, and C. Chen, "Remote sensing image scene classification based on an enhanced attention module," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 11, pp. 1926–1930, Nov. 2021.

[65] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[66] P. Deng, K. Xu, and H. Huang, "When CNNs meet vision transformer: A joint framework for remote sensing scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[67] J. Zhang, H. Zhao, and J. Li, "TRS: Transformers for remote sensing scene classification," *Remote Sens.*, vol. 13, no. 20, p. 4143, Oct. 2021.

[68] Z. Sha and J. Li, "MITformer: A multiinstance vision transformer for remote sensing scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[69] T. Peng, J. Yi, and Y. Fang, "A local–global interactive vision transformer for aerial scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.


Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:12:10 UTC from IEEE Xplore. Restrictions apply.

[70] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.
[71] B. Du, W. Xiong, J. Wu, L. Zhang, L. Zhang, and D. Tao, "Stacked convolutional denoising auto-encoders for feature representation," *IEEE Trans. Cybern.*, vol. 47, no. 4, pp. 1017–1027, Apr. 2017.
[72] D. Lin, K. Fu, Y. Wang, G. Xu, and X. Sun, "MARTA GANs: Unsupervised representation learning for remote sensing image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 11, pp. 2092–2096, Nov. 2017.
[73] Y. Wei, X. Luo, L. Hu, Y. Peng, and J. Feng, "An improved unsupervised representation learning generative adversarial network for remote sensing image scene classification," *Remote Sens. Lett.*, vol. 11, no. 6, pp. 598–607, Jun. 2020.
[74] Y. Yu, X. Li, and F. Liu, "Attention GANs: Unsupervised deep feature learning for aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 519–531, Jan. 2020.
[75] S. Ansith and A. A. Bini, "Land use classification of high resolution remote sensing images using an encoder based modified GAN architecture," *Displays*, vol. 74, Sep. 2022, Art. no. 102229.
[76] G. Cheng, X. Sun, K. Li, L. Guo, and J. Han, "Perturbation-seeking generative adversarial networks: A defense framework for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5605111.
[77] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[78] D. Guo, Y. Xia, and X. Luo, "Self-supervised GANs with similarity loss for remote sensing image scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 2508–2521, 2021.
[79] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4510–4520.
[80] N. Ma, X. Zhang, H.-T. Zheng, and J. Sun, "ShuffleNet V2: Practical guidelines for efficient CNN architecture design," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 116–131.



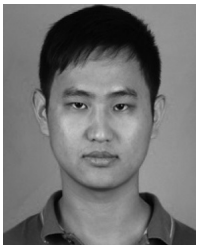
**Lingyu Si** received the M.S. degree from the University of Bristol, Bristol, U.K., in 2018. He is currently pursuing the Ph.D. degree with the University of Chinese Academy of Sciences, Beijing, China.

His research interests include transfer learning and self-supervised learning.



**Hongwei Dong** received the M.S. degree from China Agricultural University, Beijing, China, in 2018, and the Ph.D. degree from the Harbin Institute of Technology, Harbin, China, in 2022.

He is currently a Post-Doctoral Research Fellow with the Science and Technology on Integrated Information System Laboratory, Institute of Software, Chinese Academy of Sciences, Beijing. His research interests include applied mathematics, optimization methods, and machine learning.



**Wenwen Qiang** received the M.S. degree from China Agricultural University, Beijing, China, in 2018, and the Ph.D. degree from the University of Chinese Academy of Sciences, Beijing, in 2022.

He is currently a Post-Doctoral Research Fellow with the Science and Technology on Integrated Information System Laboratory, Institute of Software, Chinese Academy of Sciences, Beijing. His research interests include transfer learning, deep learning, and machine learning.



**Zeen Song** received the M.S. degree from the Technical University of Munich, Munich, Germany, in 2020. He is currently pursuing the Ph.D. degree with the University of Chinese Academy of Sciences, Beijing, China.

His research interests include self-supervised learning, contrastive learning, and machine learning.



**Bo Du** (Senior Member, IEEE) received the Ph.D. degree in photogrammetry and remote sensing from the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan, China, in 2010.

He is currently a Professor with the School of Computer Science, Wuhan University. He has more than 60 research articles published in the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, the IEEE TRANSACTIONS ON IMAGE PROCESSING, the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING (JSTARS), and the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, and 13 of them are Essential Science Indicators (ESI) hot papers or highly cited articles. His research interests include pattern recognition, hyperspectral image processing, machine learning, and signal processing.

Dr. Du was a recipient of the Distinguished Paper Award at International Joint Conference on Artificial Intelligence (IJCAI) 2018, the Best Paper Award at the IEEE Whispers 2018, the Champion Award at the IEEE Data Fusion Contest 2018, the Best Reviewer Award at the IEEE Geoscience and Remote Sensing Society (GRSS) for his service to the JSTARS in 2011, and the Association for Computing Machinery (ACM) Rising Star Awards for his academic progress in 2015. He was the Session Chair of the International Geoscience and Remote Sensing Symposium 2016 and 2018 and the fourth IEEE GRSS Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing.



**Junzhi Yu** (Fellow, IEEE) received the B.E. degree in safety engineering and the M.E. degree in precision instruments and mechanology from the North University of China, Taiyuan, China, in 1998 and 2001, respectively, and the Ph.D. degree in control theory and control engineering from the Institute of Automation, Chinese Academy of Sciences, Beijing, China, in 2003.

From 2004 to 2006, he was a Post-Doctoral Research Fellow with the Center for Systems and Control, Peking University, Beijing. In 2006, he was an Associate Professor with the Institute of Automation, Chinese Academy of Sciences, where he became a Full Professor in 2012. In 2018, he joined the College of Engineering, Peking University, as a tenured Full Professor. His research interests include intelligent robots, motion control, and intelligent mechatronic systems.



**Fuchun Sun** (Fellow, IEEE) received the Ph.D. degree in computer science and technology from Tsinghua University, Beijing, China, in 1998.

He is currently a Full Professor with the Department of Computer Science and Technology, Tsinghua University, Beijing, China. His research interests include robotic perception and cognition.

Dr. Sun was a recipient of the National Science Fund for Distinguished Young Scholars. He is an Associate Editor of many international journals, including the IEEE TRANSACTIONS ON FUZZY SYSTEMS, *Mechatronics*, and *Robotics and Autonomous Systems*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:12:10 UTC from IEEE Xplore. Restrictions apply.